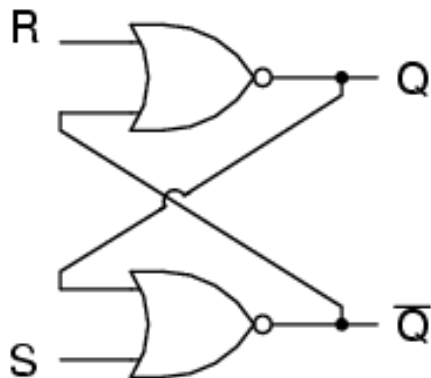


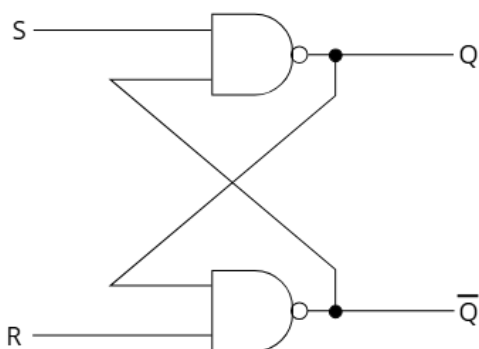
## Day 25

### SR-Latch Using NOR and NAND Gates

#### Circuit Diagram :



| S | R | Q     | $\overline{Q}$ |
|---|---|-------|----------------|
| 0 | 0 | latch | latch          |
| 0 | 1 | 0     | 1              |
| 1 | 0 | 1     | 0              |
| 1 | 1 | 0     | 0              |



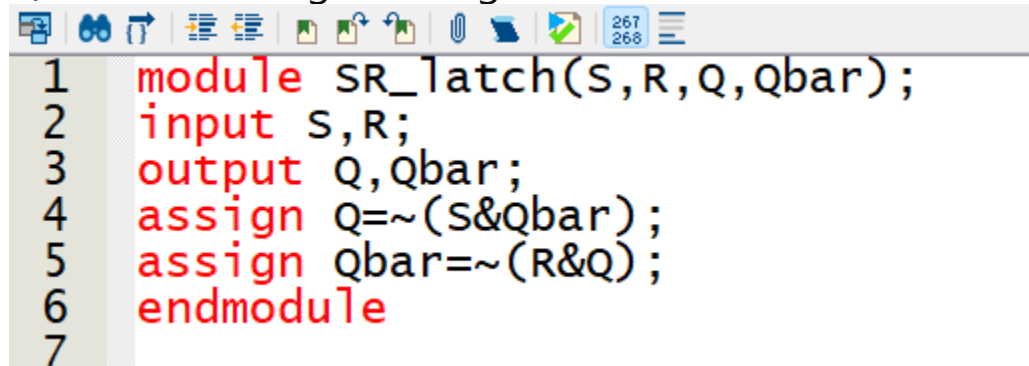
| S | R | Q | $\overline{Q}$ |                  |
|---|---|---|----------------|------------------|
| 0 | 0 | 1 | 1              | invalid          |
| 0 | 1 | 1 | 0              | set - if Q = 1   |
| 1 | 0 | 0 | 1              | reset - if Q = 0 |
| 1 | 1 | Q | $\overline{Q}$ | Hold             |

#### Verilog Code

##### A) SR-latch Using NOR gate

```
1 module SR_latch(S,R,Q,Qbar);
2   input S,R;
3   output Q,Qbar;
4   assign Q=~(R|Qbar);
5   assign Qbar=~(S|Q);
6   endmodule
7
```

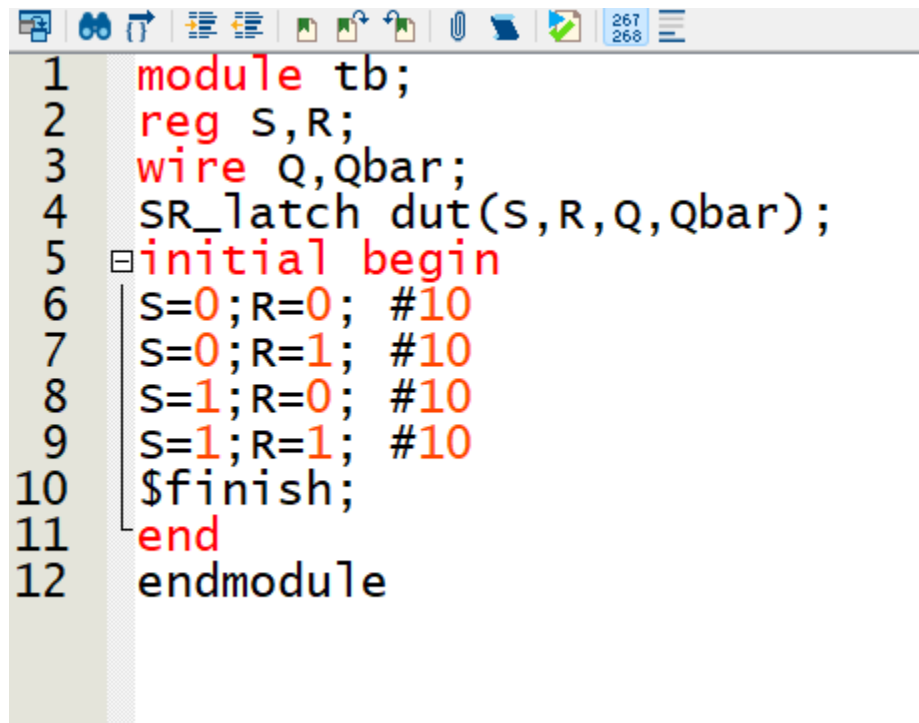
## B) SR-latch Using NAND gate



```
1 module SR_latch(S,R,Q,Qbar);
2   input S,R;
3   output Q,Qbar;
4   assign Q=~(S&Qbar);
5   assign Qbar=~(R&Q);
6   endmodule
7
```

### Testbench Code

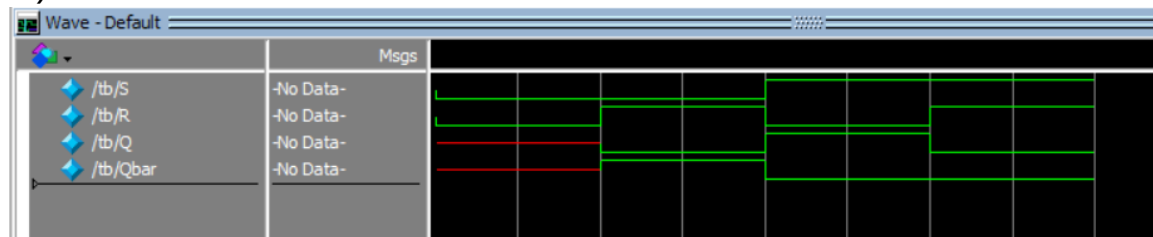
SR-latch Tb code



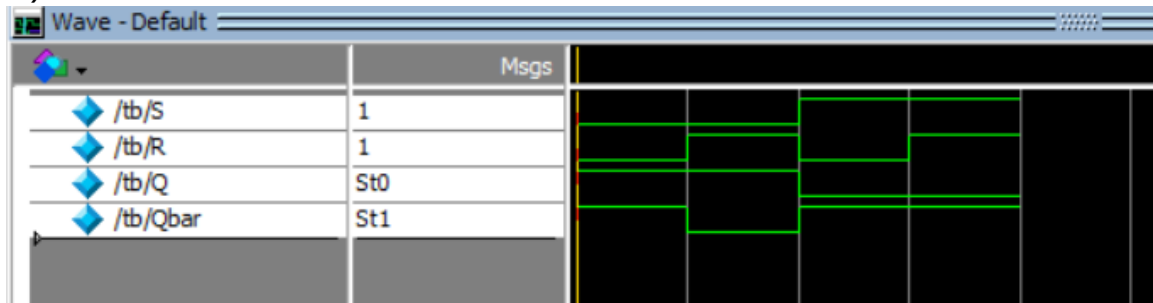
```
1 module tb;
2   reg S,R;
3   wire Q,Qbar;
4   SR_latch dut(S,R,Q,Qbar);
5   initial begin
6     S=0;R=0; #10
7     S=0;R=1; #10
8     S=1;R=0; #10
9     S=1;R=1; #10
10    $finish;
11  end
12 endmodule
```

### Waveform

A)

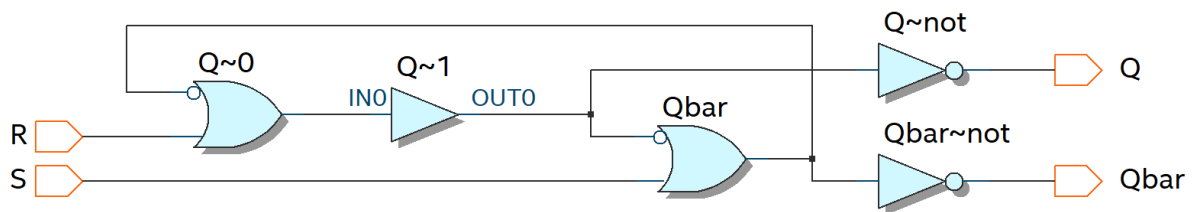


B)

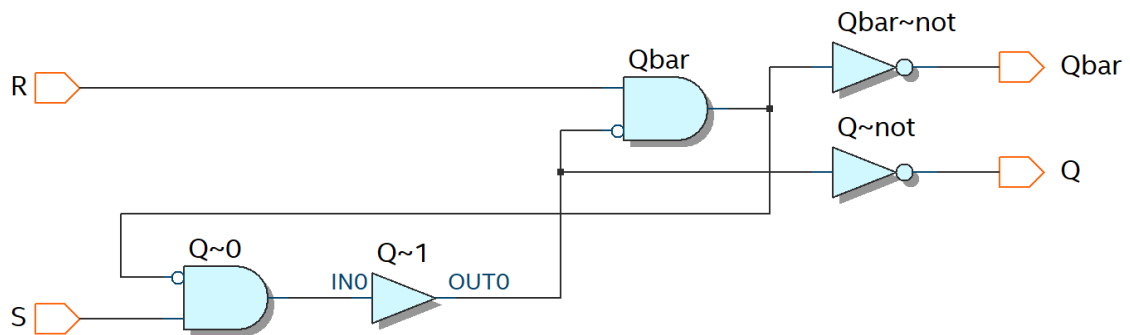


## Schematic

A)



B)



## EDA Tools Used

- IntelQuartusPrime
- ModelSim